

Euclid's Elements — Book II

Proposition 13

Formal Reconstruction — Technical Documentation

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CGJTEAMLAB

In an acute-angled triangle, the square on a side opposite an acute angle falls short of the squares on the two sides containing that angle by twice the rectangle contained by the side on which the perpendicular falls and the internal segment cut off toward the acute angle.

1 Introduction

Proposition II.13 is the acute-triangle companion to Proposition II.12. The classical configuration starts with an acute triangle ABC and drops the perpendicular from A to the carrier of BC . In the source configuration the foot D lies inside the side:

$$B - D - C, \quad AD \perp BC.$$

Euclid then combines Proposition II.7 with two applications of the Pythagorean theorem I.47.

In the lettering of the production Lean theorem, the final content relation is

$$S_{BC} + S_{AB} \simeq_{\text{ec}} S_{AC} + (R + R),$$

where R is a concrete rectangle representative contained by BC, BD . This is the formal synthetic meaning of Euclid's comparative phrase “less by twice the rectangle.” It does *not* introduce subtraction of contents, a numerical area function, multiplication of segment lengths, coordinates, or trigonometry.

The production file begins with

```
import CGJteamLab.Proposition2_12
import CGJteamLab.Proposition2_7
import CGJteamLab.Proposition47
import CGJteamLab.Proposition46
import CGJteamLab.Proposition16
```

The direct Book II theorem used for the classical content step is II.7. The import of `Proposition2_12` is architectural rather than a dependence on the obtuse-angle theorem itself: II.13 reuses general perpendicular, right-angle, square-transport, and parallelogram-cut helpers developed while formalizing II.12.

A modern law-of-cosines formula is a useful mnemonic check, but it is only an interpretive shadow. The actual proof remains in synthetic geometry and the finite scissors/equicomplementability layer.

2 Euclid's Statement and Classical Configuration

2.1 The geometric statement

Euclid's Proposition II.13 concerns an acute-angled triangle and one of its acute angles. If a perpendicular is dropped to the opposite side, the square on that opposite side is less than the two squares on the sides containing the chosen acute angle by twice the rectangle determined by the containing side on which the perpendicular falls and the internal segment cut off by the perpendicular [?].

For the current reconstruction the chosen angle is at B . The triangle is ABC , the perpendicular is AD , and the source-critical order is

$$B - D - C.$$

The corresponding geometric-content identity is

$$\text{Sq}(BC) + \text{Sq}(BA) = \text{Sq}(AC) + 2 \text{Rect}(BC, BD).$$

This displayed equation is classical control notation. The Lean theorem uses concrete figure representatives and `HilbertScissorsEquicomplementable`, not equality in a numerical area ring.

2.2 The classical construction

- (1) Start with an acute triangle ABC .
- (2) From A draw AD perpendicular to the carrier BC by I.12.
- (3) In the source configuration the foot lies internally:

$$B - D - C.$$

- (4) Apply II.7 to the divided segment BC with internal cut D .

3 Classical Proof

Euclid's local proof is short once $B - D - C$ is available. Apply II.7 to the whole segment BC cut at D :

$$\text{Sq}(BC) + \text{Sq}(BD) = 2 \text{Rect}(BC, BD) + \text{Sq}(DC).$$

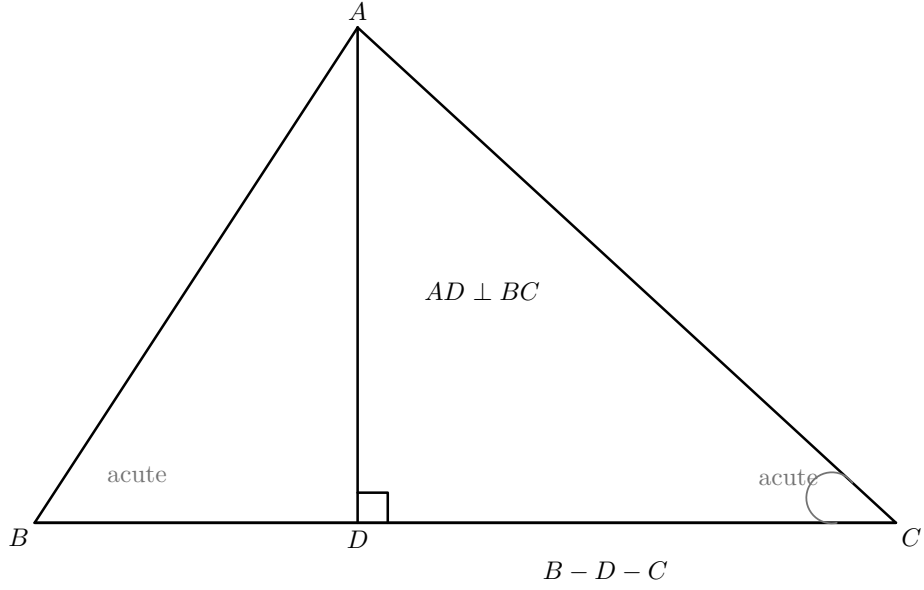


Figure 1: The classical starting configuration for Proposition II.13 in the lettering of the Lean source. The triangle ABC is acute, AD is perpendicular to the carrier BC , and the source-critical order is $B - D - C$. The square and rectangle representatives are deliberately omitted here so that the geometric input is visible before the content calculation begins.

Add the square on DA to both sides by Common Notion 2:

$$\text{Sq}(BC) + \text{Sq}(BD) + \text{Sq}(DA) = 2 \text{Rect}(BC, BD) + \text{Sq}(DC) + \text{Sq}(DA).$$

The two triangles ADB and ADC are right at D . Therefore I.47 gives

$$\text{Sq}(AB) = \text{Sq}(AD) + \text{Sq}(DB),$$

and

$$\text{Sq}(AC) = \text{Sq}(AD) + \text{Sq}(DC).$$

Substitution yields

$$\boxed{\text{Sq}(BC) + \text{Sq}(AB) = \text{Sq}(AC) + 2 \text{Rect}(BC, BD)}.$$

The local classical dependency graph is therefore

$$\begin{array}{c}
\text{acute triangle } ABC \\
\Downarrow \\
\text{I.12: } AD \perp BC \\
\Downarrow \\
B - D - C \\
\Downarrow \\
\text{II.7 on } BC \text{ cut at } D \\
S_{BC} + S_{DB} = 2R + S_{DC} \\
\Downarrow \\
\text{Common Notion 2: add } S_{DA} \\
\Downarrow \\
\text{I.47 on } ADB \quad + \quad \text{I.47 on } ADC \\
\Downarrow \\
S_{BC} + S_{AB} = S_{AC} + 2R \\
\Downarrow \\
\text{II.13.}
\end{array}$$

The formal reconstruction follows this architecture, but it must prove the internal-foot order rather than read it from the diagram, and it must manage noncanonical square and rectangle representatives explicitly.

4 The Geometric Content Identity

The production theorem does not express “less than” by an order relation on contents. It moves the square on AB to the left-hand side and exhibits the missing content positively:

$$S_{BC} + S_{AB} \simeq_{\text{ec}} S_{AC} + (R + R).$$

The two copies of R are literal formal summands. No additive inverse, area subtraction, or positivity theorem is required.

This distinction matters because the internal II.7 block is stronger than the public II.13 endpoint. The II.7 stage proves an exact finite-scissors identity

$$S_{BC} + S_{DB} \simeq_{\text{sc}} (R + R) + S_{DC},$$

represented in Lean by `HilbertScissorsEq`. The two Pythagorean relations supplied by I.47 live at the broader `HilbertScissorsEquicomplementable` level. The final theorem must therefore also conclude only equicomplementability.

5 Reusable Book II Infrastructure

5.1 Proposition II.7

The direct classical Book II dependency is II.7. The proposition-local adapter

```
proposition2_13_ii7_internal_cut
```

instantiates the production II.7 theorem with

$$(A, B, C)_{\text{II.7}} = (B, C, D)_{\text{II.13}}.$$

The exact-scissors conclusion becomes

$$S_{BC} + S_{DB} = (R + R) + S_{DC}.$$

This is precisely the normal form needed before the two I.47 substitutions.

5.2 The general parallelogram-cut constructor from II.12

The theorem

```
proposition2_12_parallelogram_cut_exists
```

was developed in the II.12 module to construct a complete internal cut of an arbitrary parallelogram. Its statement is independent of obtuse angles and of II.12’s final content identity. II.13

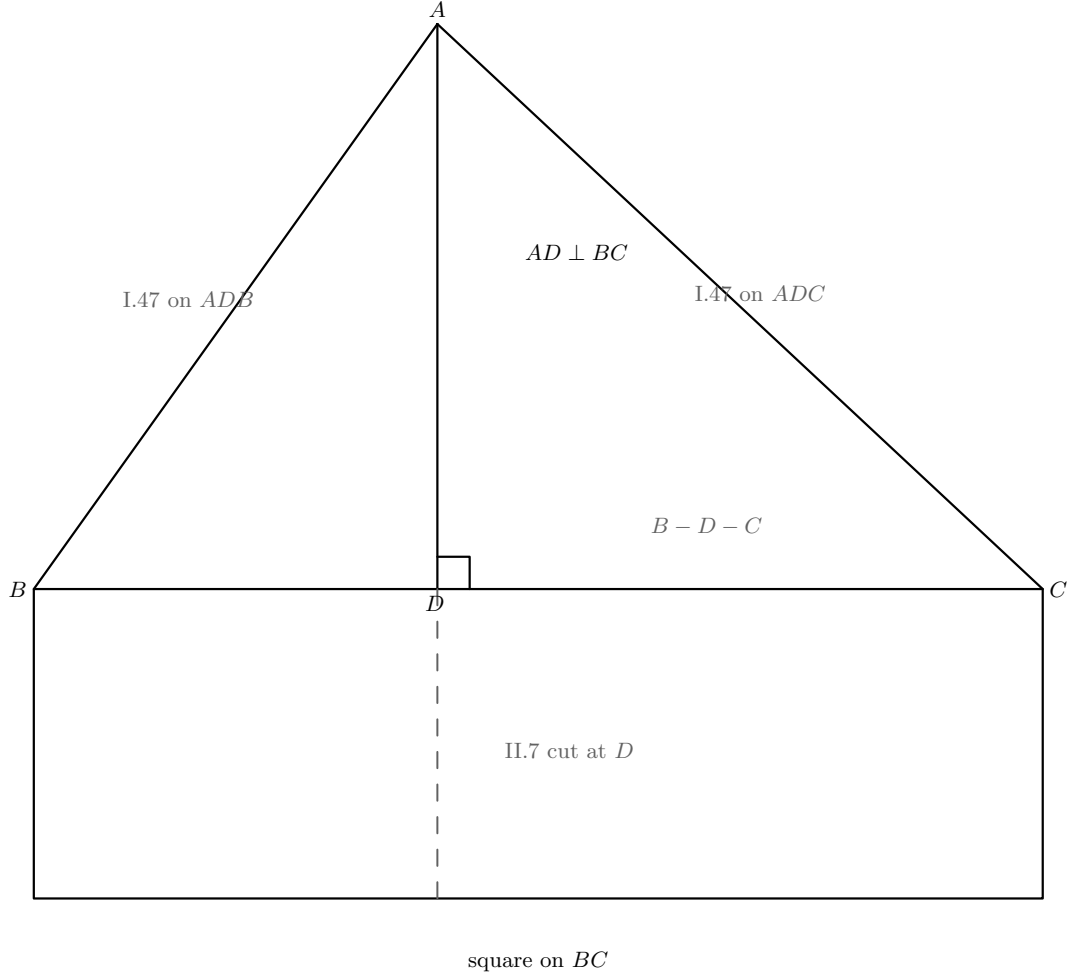


Figure 2: The content mechanism after the internal-foot order has been proved. The square on BC is cut at the point corresponding to D for the configured II.7 block, while the two right triangles ADB and ADC supply the I.47 relations. The auxiliary upper cut points, diagonal intersections, and independent rectangle witnesses required by the low-level II.7 API are omitted from the drawing; they are constructed explicitly in Lean.

reuses it three times inside `proposition2_13_ii7_exists_with_squares`: once for the square on BC , and once for each of the two configured II.3 rectangle branches hidden inside II.7.

This is an important module-DAG distinction. The Lean file imports `Proposition2_12`, but the mathematical proof of II.13 does not use the obtuse-angle theorem `euclid_proposition_2_12`. It reuses general geometric infrastructure that happens to have been established in that module.

5.3 Representative transport

The two calls to I.47 may construct different concrete squares on the same segment DA . The helper

```
proposition2_12_same_base_squares_equicomplementable
```

normalizes those representatives. After that normalization, both Pythagorean blocks literally share one DA term.

By contrast, no additional transport is needed for the DB and DC squares inside II.7. The existential II.7 constructor accepts those square representatives as inputs and reuses the exact figures returned by I.47. This proof-engineering choice removes an unnecessary layer of representative bookkeeping.

6 Proof Architecture of the Reconstruction

The formal proof naturally separates into four blocks.

6.1 Block I: synthetic acute-angle geometry

The source-level predicate is

```
def HilbertAcuteAngle
  [HilbertIncidence Geo]
  (A O B : Geo.Point) : Prop :=
  Exists fun X : Geo.Point =>
    And
      (HilbertRightAngle Geo A O X)
      (HilbertAngleLess Geo A O B A O X)

def HilbertAcuteTriangle
  [HilbertIncidence Geo]
  (A B C : Geo.Point) : Prop :=
  Not (PrimCollinear Geo A B C) /\
  HilbertAcuteAngle Geo B A C /\
  HilbertAcuteAngle Geo A B C /\
  HilbertAcuteAngle Geo A C B
```

An acute angle is therefore defined by strict synthetic angle order relative to a right angle with the same first arm. There are no degrees and no numerical angle measure.

The theorem

```
proposition2_13_acute_perpendicular_foot_between
```

starts from I.12 and proves that the foot lies between B and C . This is the main additional geometric obligation absent from Euclid’s short diagrammatic proof.

6.2 Block II: two Pythagorean identities

Once $B - D - C$ and $\angle ADB$ right are available, `proposition2_13_right_angle_other_side` transports perpendicularity across the carrier and proves $\angle ADC$ right. Then I.47 is applied to ADB and ADC .

The normalized package

```
proposition2_13_pythagorean_package_common_DA
```

returns the two content relations

$$S_{AB} \simeq_{\text{ec}} S_{DA} + S_{DB}, \quad S_{AC} \simeq_{\text{ec}} S_{DA} + S_{DC},$$

with the same concrete S_{DA} term in both formulas.

6.3 Block III: existential construction of the complete II.7 diagram

The theorem

```
proposition2_13_ii7_exists_with_squares
```

constructs all the witnesses consumed by the configured production theorem `euclid_proposition_2_7`. Its public conceptual output is only:

- a square representative on BC ;
- one representative of the rectangle contained by BC, BD ;
- the exact II.7 scissors identity.

The many internal cut points and parallelograms disappear existentially.

6.4 Block IV: pure content reassembly

The final helper

```
proposition2_13_scissors_bridge
```

contains no new geometry. Schematically it assumes

$$\begin{aligned} S_{AB} &\simeq_{\text{ec}} S_{DA} + S_{DB}, \\ S_{AC} &\simeq_{\text{ec}} S_{DA} + S_{DC}, \\ S_{BC} + S_{DB} &\simeq_{\text{sc}} (R + R) + S_{DC}, \end{aligned}$$

and proves

$$S_{BC} + S_{AB} \simeq_{\text{ec}} S_{AC} + (R + R).$$

The proof uses reflexivity, compatibility with formal addition, conversion of exact scissors equality to equicomplementability, symmetry, transitivity, and associative-commutative regrouping. It uses no cancellation theorem.

7 Lean Formalization

7.1 The proposition-local theorem chain

The production module contains the following declarations:

```

proposition2_13_right_angle_other_side
proposition2_13_two_pythagoras
proposition2_13_pythagorean_package_common_DA
proposition2_13_ii7_internal_cut
proposition2_13_scissors_bridge
HilbertAcuteAngle
HilbertAcuteTriangle
proposition2_13_acute_not_right
proposition2_13_acute_perpendicular_foot_between
proposition2_13_acute_triangle_perpendicular_foot_between
proposition2_13_ii7_exists_with_squares
euclid_proposition_2_13

```

The source order reflects the incremental implementation. The mathematical DAG is more naturally read as

acute-foot geometry $\longrightarrow$ two I.47 blocks $\longrightarrow$ configured II.7 block $\longrightarrow$ final scissors bridge.

7.2 The public theorem

The exact public signature is:

```

theorem euclid_proposition_2_13
  [HilbertIncidence Geo]
  [HilbertEuclideanPlane Geo]

  (A B C : Geo.Point)
  (hAcuteTriangle : HilbertAcuteTriangle Geo A B C) :

  exists D : Geo.Point,
  exists AB0 AB1 AC0 AC1 BC0 BC1 : Geo.Point,
  exists T0 T1 T2 T3 : Geo.Point,

  Geo.Between B D C /\
  HilbertRightAngle Geo A D B /\

  IsSquare Geo A B AB1 AB0 /\
  IsSquare Geo A C AC1 AC0 /\
  IsSquare Geo B C BC1 BC0 /\

  IsRectangleContainedBy Geo
    T0 T1 T2 T3 B C B D /\

  HilbertScissorsEquicomplementtable Geo
    (hilbertParallelogramTerm Geo B C BC1 BC0 +
     hilbertParallelogramTerm Geo A B AB1 AB0)
    (hilbertParallelogramTerm Geo A C AC1 AC0 +
     (hilbertParallelogramTerm Geo T0 T1 T2 T3 +
      hilbertParallelogramTerm Geo T0 T1 T2 T3))

```

The public theorem therefore exposes only the geometrically meaningful foot, three target squares, one target rectangle representative, and the final content relation. All intermediate squares, cross rectangles, cut points, diagonal intersections, and parallelograms remain hidden witnesses.

8 Relationship with Earlier Book II Propositions

8.1 The classical dependency is II.7

Euclid’s local Book II dependency is II.7. The source proof uses its arbitrary internal-cut identity on BC divided at D . It does not use II.12.

8.2 The Lean module also imports the previous production module

The formal module imports `Proposition2_12` because that file contains general reusable helpers obtained during the previous construction:

- perpendicular-foot construction in a usable incidence form;
- right-angle arm swap and collinear carrier transport;
- same-base square equicomplementability;
- the generic parallelogram internal-cut constructor.

These are library-level implementation dependencies. They should not be reported as a classical theorem edge

$$\text{II.12} \longrightarrow \text{II.13}.$$

The correct distinction is:

classical Book II DAG: II.7

Lean module DAG: `Proposition2_12` + `Proposition2_7` + Book I modules.

9 Relationship with Book I Infrastructure

The main Book I roles are:

- I.12 constructs the perpendicular from A to the carrier BC ;
- I.16 excludes both external-foot cases and forces $B - D - C$;
- I.47 supplies the two Pythagorean content relations;
- I.46 constructs the square on BC used by the existential II.7 block.

The proof also relies on established Hilbert infrastructure for strict betweenness, betweenness trichotomy, noncollinearity transport, strict angle order, congruence of all right angles, right-angle transport, parallel uniqueness, parallelogram construction, Pasch, rectangle existence, and the scissors/equicomplementability calculus.

10 Formalization Notes and Hidden Geometric Data

10.1 The production wrapper assumes an acute triangle

The final theorem takes `HilbertAcuteTriangle`, not merely a single local acute-angle hypothesis. This matches Euclid’s acute-triangle setting and is mathematically useful: the angles at both B and C are available to rule out the two possible external locations of the perpendicular foot.

The helper proving the foot position actually needs only the noncollinearity of ABC together with the acute angles at B and C . The acute angle at A is retained by the public wrapper because the proposition is documented as the acute-triangle case rather than as a maximally weakened local lemma.

10.2 The internal order is proved, not assumed

I.12 gives a perpendicular to the *carrier* of BC . It does not by itself say where the foot lies. Lean first excludes $D = B$ and $D = C$: either equality would make the corresponding triangle angle right, contradicting acute angle order.

For the remaining collinear triple B, D, C , betweenness trichotomy leaves three strict cases. The desired branch is $B - D - C$. The two other branches are synthetically contradictory.



Figure 3: The two external-foot branches eliminated by the Lean proof. If $D - B - C$, I.16 in triangle ADB makes the right angle ADB strictly smaller than the exterior angle ABC , contradicting the acute angle at B . If $B - C - D$, the reversed order $D - C - B$ and I.16 in triangle ADC make the right angle ADC strictly smaller than ACB , contradicting the acute angle at C . The drawings display the *reductio* configurations; they are not additional hypotheses of the final theorem.

10.3 Acuteness is strict angle order, not a number

The contradiction just described never uses 90° . A right-angle witness and `HilbertAngleLess` are enough. Congruence of right angles and transport of strict angle order produce an impossible cycle

$$\alpha < \beta < \alpha.$$

Irreflexivity of strict angle order closes each external branch.

10.4 Perpendicularity must be transported to the other ray

I.12 yields one right-angle orientation. The second Pythagorean triangle needs $\angle ADC$ right as well. The helper `proposition2_13_right_angle_other_side` uses the collinearity $B - D - C$, swaps the arms of the right angle, transports one arm along the carrier, and swaps back. This is the formal form of the classical statement that AD is perpendicular to the whole line BC , not merely to one ray.

10.5 The common DA square is a representation normalization

I.47 is existential at the square-representation level. Two calls can choose different vertices for the square on DA . The theorem `proposition2_13_pythagorean_package_common_DA` transports the second representative to the first at the content level. No claim is made that square vertices are definitionally unique.

10.6 The II.7 construction reuses the DB and DC squares literally

The existential II.7 wrapper is deliberately parametrized by the square representatives on DB and DC . Passing the I.47 representatives directly makes the later terms syntactically identical. This is a proof-engineering normalization, not an additional geometric restriction.

10.7 The low-level II.7 witnesses are genuine cut geometry

The configured II.7 API requires more than the order $B - D - C$. Internally it needs a square on BC , upper cut points, diagonal intersections, two parallelogram pieces for each cut, contained-by rectangle representatives, and a cross rectangle. The generic constructor inherited from the II.12 module builds these witnesses with parallel geometry and Pasch.

They do not appear in the public theorem, but they certify that the exact `HilbertScissorsEq` used in the middle of the proof is backed by an actual finite geometric cut.

10.8 “Less by” is not content subtraction

The final relation is intentionally oriented as

$$S_{BC} + S_{AB} \simeq_{\text{ec}} S_{AC} + R + R.$$

Thus the content by which the left square is “less” is exhibited as a positive summand on the opposite side. The theorem does not define $S_{AC} - S_{BC}$ and does not require an order relation or additive inverses on scissors terms.

10.9 `ac_rfl` is only formal regrouping

The pure scissors bridge uses `ac_rfl` to expose the Pythagorean blocks inside larger multiset sums. This proves only equality under associativity and commutativity of formal addition. It does not replace any geometric content step: II.7 and both I.47 relations have already been established before the regrouping occurs.

11 Axiomatic and Semantic Status

The production theorem introduces no new proposition-specific geometric axiom and no new content axiom. It contains no `sorry` or `admit`. The production module and the full project build were both reported clean.

The transitive audit

```
#print axioms Geometry.proposition2_13_acute_perpendicular_foot_between
#print axioms Geometry.proposition2_13_ii7_exists_with_squares
#print axioms Geometry.euclid_proposition_2_13
```

reported the same dependencies for all three declarations:

```
[propext,
 Classical.choice,
 Geometry.bookZero_nullSegment1,
 Quot.sound]
```

These are inherited project dependencies. The precise claim is therefore: II.13 adds no new axiom and no `sorry`; it is not literally axiom-free.

The semantic layers are

$$\begin{array}{c}
 \text{incidence/order/congruence} + \text{Euclidean parallel geometry} \\
 \Downarrow \\
 \text{synthetic acute-angle order} + \text{I.12 perpendicular construction} \\
 \Downarrow \\
 \text{I.16 proof of } B - D - C \\
 \Downarrow \\
 \text{two right triangles and two I.47 equicomplementabilities} \\
 \Downarrow \\
 \text{configured exact II.7 scissors identity} \\
 \Downarrow \\
 \text{equicomplementability addition and substitution} \\
 \Downarrow \\
 \text{II.13 additive equal-content conclusion.}
 \end{array}$$

No numerical area function is used. No segment multiplication is used. No coordinate model, trigonometric function, numerical angle measure, content order, or subtraction operation is used.

12 Source Interpretation

Hartshorne’s equal-content framework is the appropriate semantic control for the final equality of figures. It allows the Book II identities to be read as synthetic content statements rather than numerical area equations [?]. II.13 is especially useful here because the phrase “less by” can tempt a modern reader to introduce subtraction. The formal theorem instead exhibits an additive equal-content relation.

Hilbert supplies the architectural distinction between exact finite equidecomposition, the broader equicomplementability relation, and the later arithmetic of segments [?]. II.13 crosses from an

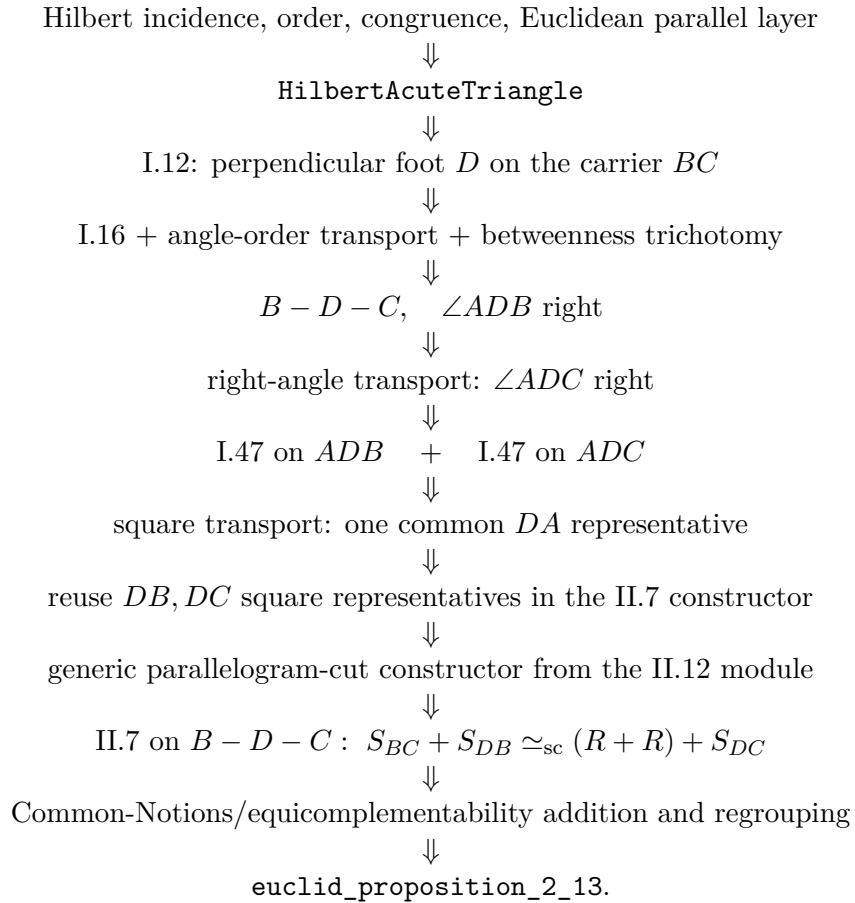
internal exact II.7 dissection to a public equicomplementability statement because the I.47 blocks already live at that level; it does not cross into Hilbert’s later segment multiplication.

Greenberg remains a control source for Hilbert-style order, congruence, perpendicular and exterior-angle geometry, and for the warning that equal content is not numerical area. The particular finite scissors calculus is project infrastructure rather than a theorem attributed to Greenberg [?].

Borsuk–Szmielw are relevant to the incidence/order/congruence and representation shell: the location of the perpendicular foot, parallelogram cuts, parallel transport, and Pasch-type intersection belong on that geometric side. The current source audit does not attribute the project’s `HilbertScissorsEquicomplementable` calculus to their text [?].

13 Dependency Summary

The actual production chain is



The implementation checkpoint is:

New geometric axiom in II.13:	no
New content axiom in II.13:	no
New proposition-local helper theorem:	yes
New public acute-angle / acute-triangle definitions:	yes
New helper promoted to Interface / Book Zero / Scissors:	no
New <code>HilbertRectangle</code> theorem:	no
Directly reuses earlier Book II theorem:	yes, II.7
Reuses general helper infrastructure from II.12 module:	yes
Conclusion is exact <code>HilbertScissorsEq</code> :	no
Conclusion is equal-content / equicomplementability:	yes
Uses exact scissors internally:	yes, through II.7
Uses content cancellation:	no
Uses numerical area:	no
Uses segment multiplication:	no
Uses numerical angle measure / trigonometry:	no
Production source compile:	clean (user verified)
Full project build:	clean (user verified)
Proposition-local <code>sorry/admit</code> :	no
Transitive <code>#print axioms</code> :	<code>propext</code> , <code>Classical.choice</code> , <code>bookZero_nullSegment1</code> , <code>Quot.sound</code>

14 Conclusion

Proposition II.13 preserves Euclid’s compact top-level proof:

$$\boxed{\text{I.12} \longrightarrow \text{II.7} \longrightarrow \text{add } DA^2 \longrightarrow \text{I.47 twice} \longrightarrow \text{II.13.}}$$

The formal development is longer because two diagrammatic assumptions must become theorems or explicit witness packages.

First, Lean proves that the perpendicular from A meets the *segment* BC , not merely its carrier. The two external positions are eliminated synthetically by I.16 and strict angle order. Second, the configured II.7 API requires a complete geometric cut diagram, so the proof constructs the necessary parallelogram and intersection witnesses before invoking the exact scissors identity.

Once those obligations are discharged, the final content argument is very close to Euclid’s proof. The theorem states

$$\boxed{S_{BC} + S_{AB} \simeq_{\text{ec}} S_{AC} + (R + R)}$$

for a rectangle representative R contained by BC, BD . This is a synthetic, positive equal-content interpretation of “less by twice the rectangle,” with no subtraction, numerical area, segment field, coordinates, cosine, or numerical angle measure.